

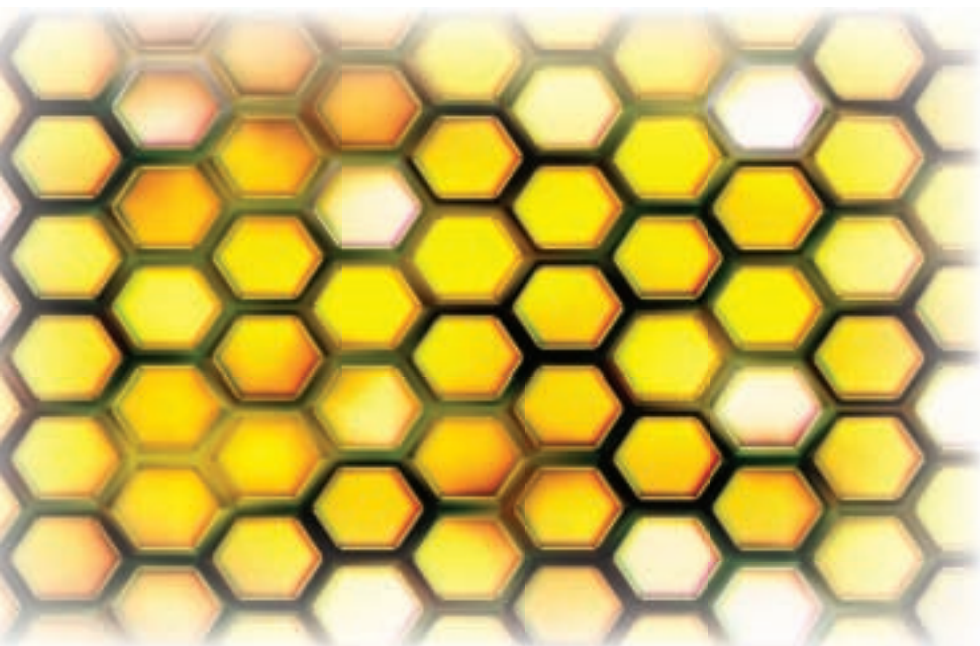
Need battery?

Zen Mobile has introduced the M25 phone that claims to have a battery life of 72 hours

Laser zapping

US scientists at Wake Forest University in North Carolina have demonstrated the concept of laser zapping to treat mice with breast tumours to heal them

THE REAL NANOS



Photos.com

"The Next Big Thing is Really Small"
-Jack Uldrich

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Nano as a prefix implies a scale of 10^{-9} and nanotechnology is the science and study of manipulating matter at the atomic or molecular scale and building materials with at least one dimension under 100 nanometres. Although the first reference towards nanotechnology was given by legendary Richard Feynman in 1959, carbon nanotubes were only introduced in 1991 and since then have become the focal point of the scientific community. From super computers to space elevators, their use have been envisioned in everything, so we will do a reality check today and discuss their current use, ongoing research and future applications.

If you peer down with a good microscope you'll observe, that at the atomic level, carbon nanotubes resemble your nylon clothing line. Only instead of nylon, the tube is made

from a hexagonal lattice of carbon atoms. The tubes are in the order of nanometres in diameter. Their structural strength makes them an ideal composite material, while its electrical conductivity makes it suitable for deployment as a semiconductor alternative.

Current application Copper Replacement

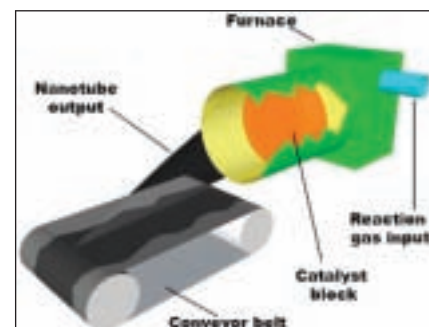
Long Fibres

Today nanotubes are commonly produced in segments about 10,000 nanometres long, forming a black powder. Nanocomp Technologies of Concord manufacture lightweight antennas, cables, and electromagnetic interference shields using this same material for the military and aerospace markets.

The heart of Nanocomp Technologies' manufacturing process is a long, cylindrical furnace. A reaction gas comprising carbon-based feedstock which is the raw material is fed into the furnace. The gas is swept along the furnace until it meets a heated, porous block coated with a catalyst.

As it passes through the pores, the gas is forced to be in contact with the catalyst, and carbon nanotubes are formed. A temperature gradient between the porous block and the open end of the furnace supports the tubes as they grow, absorbing material from the reaction gas. Eventually, nanotubes leave the cylindrical furnace, looking like a black mass of tube-shaped cotton candy. A rotating anchor or conveyor belt captures the nanotubes as they emerge, turning them into yarns and sheets.

They are also producing sheets of carbon nanotubes that measure 3 x 6 feet and have announced plans to produce slabs with an area of 100 square feet. These sheets will be used primarily in as electrical



LongYarn Image Courtesy: Stephen Cass

conductors in planes and satellites to replace copper wire and save weight, and hence increase fuel efficiency. The resulting material can be a valuable addition to such applications such as electromagnetic interference (EMI) shielding, electrical conductors, thermal dissipation solutions, lightning protection and advanced structural composites. Another interesting use due to structural strength of the material is as a substitute for kevlar as body armour.

Agriculture

Carbon nanotubes are being used to accelerate the growth in size, quantity and speed, of vegetables like tomatoes. In September 2009, at the Little Rock Nanotechnology Center, housed at the University of Arkansas, used carbon nanotubes to increase the germination rate of tomatoes. Exposure to carbon nanotubes caused the seedlings to become stronger and grow more quickly. Carbon nanotubes become a part of the seed by penetrating through the